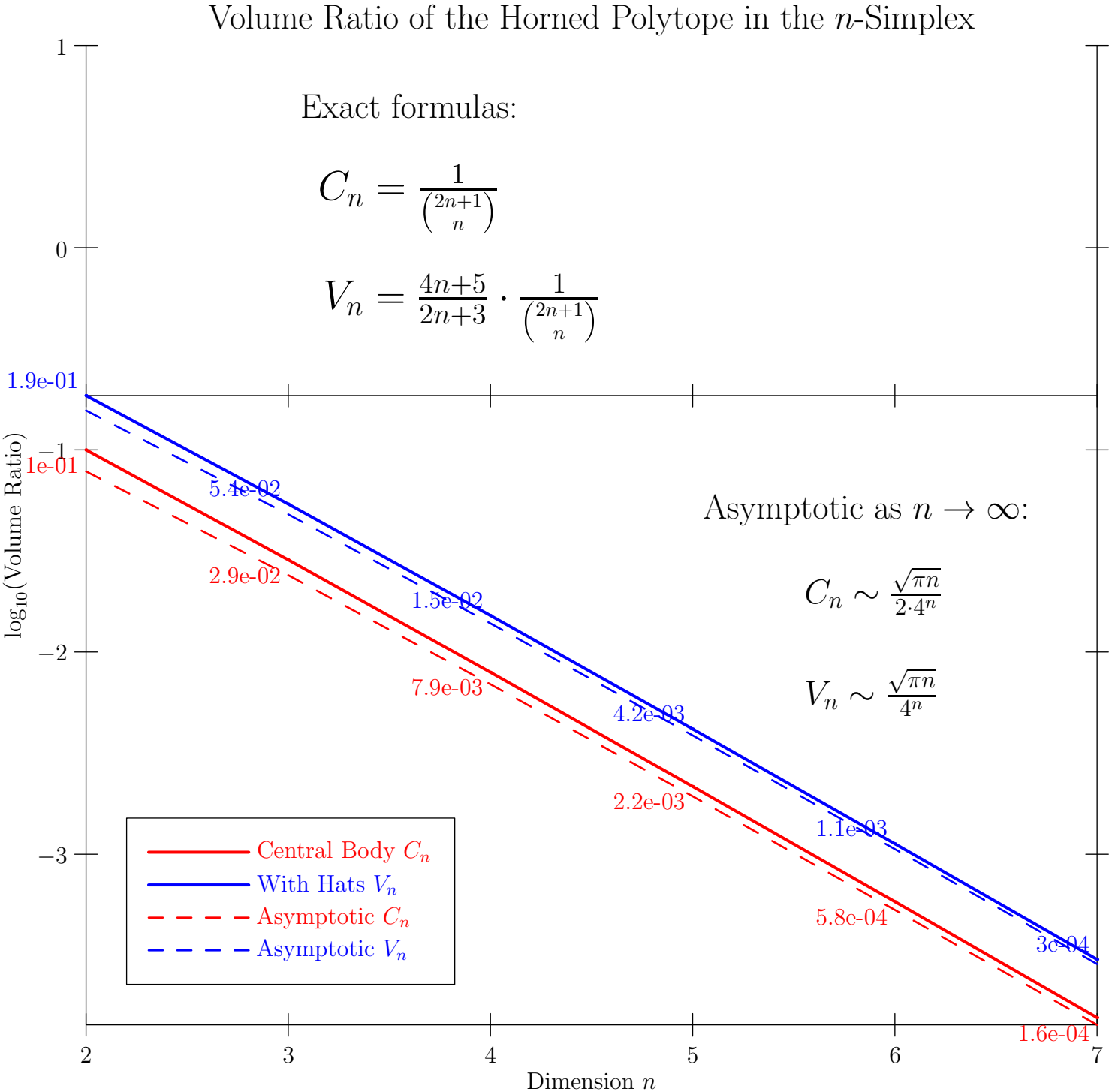
Volume Ratio of the Horned Polytope in the n-Simplex
Exact formulas:
$C_n = \frac{1}{\binom{2n+1}{n}}$
$V_n = \frac{4n+5}{2n+3} \cdot \frac{1}{\binom{2n+1}{n}}$
Asymptotic as $n \to \infty$:
$C_n \sim \frac{\sqrt{\pi n}}{2 \cdot 4^n}$
$V_n \sim \frac{\sqrt{\pi n}}{4^n}$
$\log_{10}$(Volume Ratio)
Dimension n
1
0
−1
−2
−3
2
3
4
5
6
7
1.9e-01
1e-01
5.4e-02
2.9e-02
1.5e-02
7.9e-03
4.2e-03
2.2e-03
1.1e-03
5.8e-04
3e-04
1.6e-04
Central Body C_n
With Hats V_n
Asymptotic C_n
Asymptotic V_n